What is a Hopf Algebra and why should we care?

AFRICAN MATHEMATICS SEMINAR

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1. Definitions

k is an algebraically closed field of characteristic 0.
$\otimes$ will always mean $\otimes_k$, tensor product over k.

Definition

An (associative) k-algebra is a k-vector space R which is also a ring with an identity element 1, with the scalars $\{\lambda 1 : \lambda \in k\}$ in the centre of R;

Definition

A coalgebra *is a k-vector space C with linear maps $\Delta : C \longrightarrow C \otimes C$ and $\varepsilon : C \longrightarrow k$ such that*

Cocommutative classification theorem in char 0

For cocommutative Hopf algebras in char 0, group algebras and enveloping algebras are almost the complete story:

Theorem

(Cartier-Gabriel-Kostant (1968)) Let H be a cocommutative Hopf k-algebra. Recall that we're assuming that k is algebraically closed and has characteristic 0.

Noncommutative and non-cocommutative examp

Maybe every Hopf algebra is commutative or cocommutative? NO!!!

Example

(Sweedler) Let H be the 4-dimensional vector space

$$H = k1 + kx + kg + kxg,$$

multiplication given by $g^2 = 1$, $x^2 = 0$, $xg = -gx$.

Why should we care? I. Tensoring representations(2)

Every Hopf algebra has a special 1-dimensional module, the trivial module k, given by

$$hv := \varepsilon(h)v, \ h \in H, v \in k;$$

from the Hopf algebra axioms we get $V \otimes k \cong V \cong k \otimes V$ for every H-representation V. And coassociativity of Δ ensures that $(V \otimes W) \otimes Y \cong V \otimes (W \otimes Y)$.

So we get a structure of monoid on the category of H-representations. This leads us to the concept of a tensor category - a very important current research area.

References

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